INTERNATIONAL STANDARD

ISO 14064-1

Second edition
2018-12

Greenhouse gases —

Part 1: Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals

Gaz à effet de serre —

Partie 1: Spécifications et lignes directrices, au niveau des organismes, pour la quantification et la déclaration des émissions et des suppressions des gaz à effet de serre

Reference number
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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 207, *Environmental management*, Subcommittee SC 7, *Greenhouse gas management and related activities*.

This second edition cancels and replaces the first edition (ISO 14064-1:2006), which has been technically revised.

The main changes compared to the previous edition are as follows.

- A new approach has been introduced to reporting boundaries, facilitating the inclusion and expansion of indirect emissions. This change is a response to a growing number of organizations that are recognizing the importance and significance of indirect emissions and are developing GHG inventories that include more types of indirect emissions across the value chain.

- The GHG emissions category “other indirect GHG emissions” has been renamed “indirect GHG emissions.” Requirements and guidance have been provided for classification of indirect GHG emissions into five specific categories. “Operational boundaries” has been renamed “reporting boundaries” for clarification and simplicity.

- New requirements and guidance for GHG quantification and reporting of specific items, such as the treatment of biogenic carbon and GHG emissions related to electricity, have been added for clarification.

A list of all parts in the ISO 14064 series can be found on the ISO website.

This document is the generic standard for the quantification and reporting of greenhouse gas emission and removals at an organizational level.

Any feedback or questions on this document should be directed to the user’s national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

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Introduction

0.1 Background

Climate change arising from anthropogenic activity has been identified as one of the greatest challenges facing the world and will continue to affect business and citizens over future decades.

Climate change has implications for both human and natural systems and could lead to significant impacts on resource availability, economic activity and human wellbeing. In response, international, regional, national and local initiatives are being developed and implemented by public and private sectors to mitigate greenhouse gas (GHG) concentrations in the Earth's atmosphere, as well as to facilitate adaptation to climate change.

There is a need for an effective and progressive response to the urgent threat of climate change on the basis of the best available scientific knowledge. ISO produces documents that support the transformation of scientific knowledge into tools that will help address climate change.

GHG initiatives on mitigation rely on the quantification, monitoring, reporting and verification of GHG emissions and/or removals.

The ISO 14060 family provides clarity and consistency for quantifying, monitoring, reporting and validating or verifying GHG emissions and removals to support sustainable development through a low-carbon economy and to benefit organizations, project proponents and interested parties worldwide. Specifically, the use of the ISO 14060 family:

- enhances the environmental integrity of GHG quantification;
- enhances the credibility, consistency and transparency of GHG quantification, monitoring, reporting, verification and validation;
- facilitates the development and implementation of GHG management strategies and plans;
- facilitates the development and implementation of mitigation actions through emission reductions or removal enhancements;
- facilitates the ability to track performance and progress in the reduction of GHG emissions and/or increase in GHG removals.

Applications of the ISO 14060 family include:

- corporate decisions, such as identifying emission reduction opportunities and increasing profitability by reducing energy consumption;
- risks and opportunities management, such as climate-related risks, including financial, regulatory, supply chain, product and customer, litigation, reputational risks and its opportunity for business (e.g. new market, new business model);
- voluntary initiatives, such as participation in voluntary GHG registries or sustainability reporting initiatives;
- GHG markets, such as the buying and selling of GHG allowances or credits;
- regulatory/government GHG programmes, such as credit for early action, agreements or national and local reporting initiatives.

This document details principles and requirements for designing, developing, managing and reporting organization-level GHG inventories. It includes requirements for determining GHG emission and removal boundaries, quantifying an organization's GHG emissions and removals, and identifying specific company actions or activities aimed at improving GHG management. It also includes requirements and guidance on inventory quality management, reporting, internal auditing and the organization's responsibilities in verification activities.

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ISO 14064-2 details principles and requirements for determining baselines, and monitoring, quantifying and reporting of project emissions. It focuses on GHG projects or project-based activities specifically designed to reduce GHG emissions and/or enhance GHG removals. It provides the basis for GHG projects to be verified and validated.

ISO 14064-3 details requirements for verifying GHG statements related to GHG inventories, GHG projects, and carbon footprints of products. It describes the process for verification or validation, including verification or validation planning, assessment procedures, and the evaluation of organizational, project and product GHG statements.

ISO 14065 defines requirements for bodies that validate and verify GHG statements. Its requirements cover impartiality, competence, communication, validation and verification processes, appeals, complaints and the management system of validation and verification bodies. It can be used as a basis for accreditation and other forms of recognition in relation to the impartiality, competence and consistency of validation and verification bodies.

ISO 14066 specifies competence requirements for validation teams and verification teams. It includes principles and specifies competence requirements based on the tasks that validation teams or verification teams have to be able to perform.

ISO 14067 defines the principles, requirements and guidelines for the quantification of the carbon footprint of products. The aim of ISO 14067 is to quantify GHG emissions associated with the life cycle stages of a product, beginning with resource extraction and raw material sourcing and extending through the production, use and end-of-life phases of the product.

ISO/TR 14069 assists users in the application of this document, providing guidelines and examples for improving transparency in the quantification of emissions and their reporting. It does not provide additional guidance to this document.

Figure 1 illustrates the relationship among the ISO 14060 family of GHG standards.

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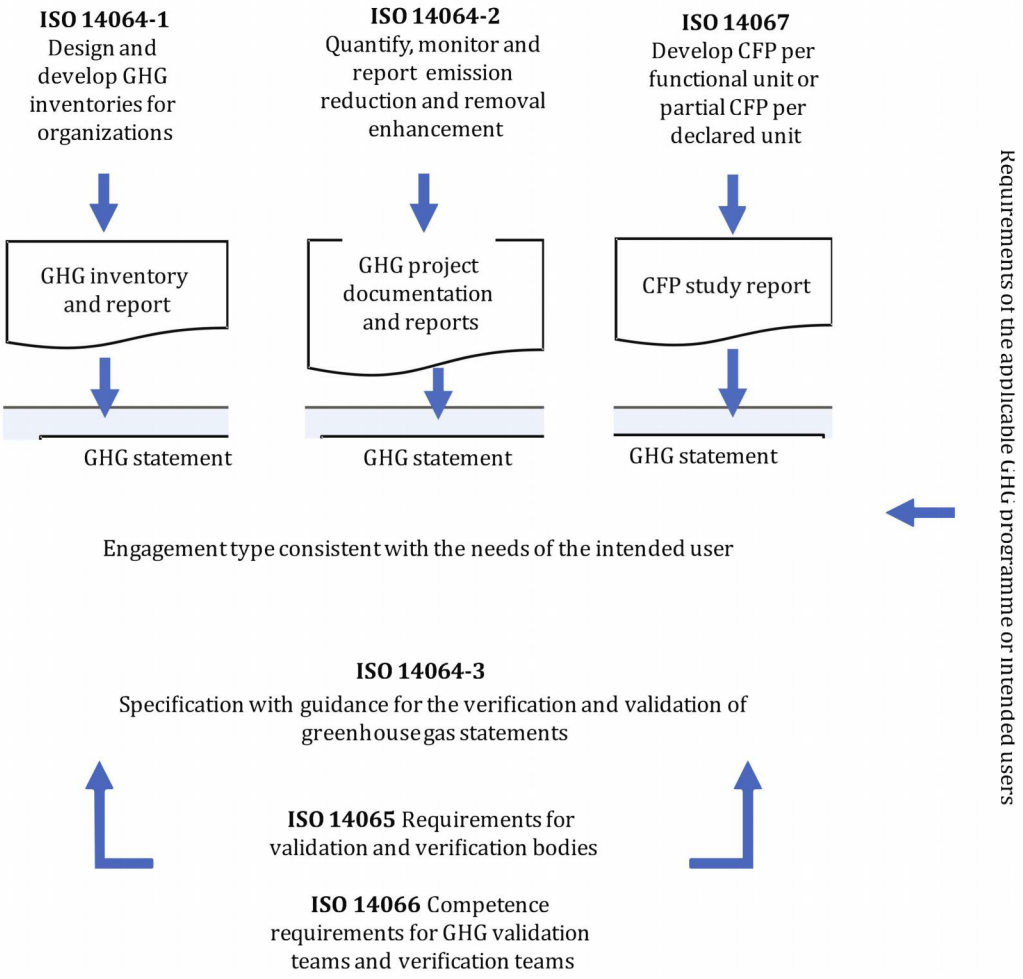


Figure 1 — Relationship among the ISO 14060 family of GHG standards

0.2 Base GHG quantification concepts used by this document

This document incorporates many key concepts developed over a number of years. References listed in the Bibliography provide (examples of) additional guidance on these concepts.

0.3 Significance of the terms "document", "explain" and "justify" in this document

Some clauses require users of this document to document, explain and justify the use of certain approaches or decisions taken.

Document involves capturing and storing the pertinent information in writing.

Explain involves two additional criteria:

a) describe how approaches were used or decisions taken, and

b) describe why approaches were chosen or decisions made.

Justify involves an additional third and fourth criteria:

c) explain why alternative approaches were not chosen, and

d) provide supporting data or analysis.

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Greenhouse gases —

Part 1:
Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals

1 Scope

This document specifies principles and requirements at the organization level for the quantification and reporting of greenhouse gas (GHG) emissions and removals. It includes requirements for the design, development, management, reporting and verification of an organization's GHG inventory.

The ISO 14064 series is GHG programme neutral. If a GHG programme is applicable, requirements of that GHG programme are additional to the requirements of the ISO 14064 series.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

ISO Online browsing platform: available at https://www.iso.org/obp

IEC Electropedia: available at http://www.electropedia.org/

3.1 Terms relating to greenhouse gases

3.1.1
greenhouse gas
GHG
gaseous constituent of the atmosphere, both natural and anthropogenic, that absorbs and emits radiation at specific wavelengths within the spectrum of infrared radiation emitted by the Earth's surface, the atmosphere and clouds

Note 1 to entry: For a list of GHGs, see the latest Intergovernmental Panel on Climate Change (IPCC) Assessment Report.

Note 2 to entry: Water vapour and ozone are anthropogenic as well as natural GHGs, but are not included as recognized GHGs due to difficulties, in most cases, in isolating the human-induced component of global warming attributable to their presence in the atmosphere.

3.1.2
greenhouse gas source
GHG source
process that releases a *GHG* (3.1.1) into the atmosphere

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3.1.3
greenhouse gas sink
GHG sink
process that removes a *GHG* (3.1.1) from the atmosphere

3.1.4
greenhouse gas reservoir
GHG reservoir
component, other than the atmosphere, that has the capacity to accumulate *GHGs* (3.1.1), and to store and release them

Note 1 to entry: Oceans, soils and forests are examples of components that can act as reservoirs.

Note 2 to entry: GHG capture and storage is one of the processes that results in a GHG reservoir.

3.1.5
greenhouse gas emission
GHG emission
release of a *GHG* (3.1.1) into the atmosphere

3.1.6
greenhouse gas removal
GHG removal
withdrawal of a *GHG* (3.1.1) from the atmosphere by *GHG sinks* (3.1.3)

3.1.7
greenhouse gas emission factor
GHG emission factor
coefficient relating *GHG activity data* (3.2.1) with the *GHG emission* (3.1.5)

Note 1 to entry: A GHG emission factor could include an oxidation component.

3.1.8
greenhouse gas removal factor
GHG removal factor
coefficient relating *GHG activity data* (3.2.1) with the *GHG removal* (3.1.6)

Note 1 to entry: A GHG removal factor could include an oxidation component.

3.1.9
direct greenhouse gas emission
direct GHG emission
GHG emission (3.1.5) from *GHG sources* (3.1.2) owned or controlled by the *organization* (3.4.2)

Note 1 to entry: This document uses the concepts of equity share or control (financial or operational control) to establish organizational boundaries.

3.1.10
direct greenhouse gas removal
direct GHG removal
GHG removal (3.1.6) from *GHG sinks* (3.1.3) owned or controlled by the *organization* (3.4.2)

3.1.11
indirect greenhouse gas emission
indirect GHG emission
GHG emission (3.1.5) that is a consequence of an *organization's* (3.4.2) operations and activities, but that arises from *GHG sources* (3.1.2) that are not owned or controlled by the organization

Note 1 to entry: These emissions occur generally in the upstream and/or downstream chain.

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3.1.12
global warming potential
GWP
index, based on radiative properties of *GHGs* (3.1.1), measuring the radiative forcing following a pulse emission of a unit mass of a given GHG in the present-day atmosphere integrated over a chosen time horizon, relative to that of carbon dioxide (CO_2)

3.1.13
carbon dioxide equivalent
CO_2e
unit for comparing the radiative forcing of a *GHG* (3.1.1) to that of carbon dioxide

Note 1 to entry: The carbon dioxide equivalent is calculated using the mass of a given GHG multiplied by its *global warming potential* (3.1.12).

3.2 Terms relating to the GHG inventory process

3.2.1
greenhouse gas activity data
GHG activity data
quantitative measure of activity that results in a *GHG emission* (3.1.5) or *GHG removal* (3.1.6)

EXAMPLE Amount of energy, fuels or electricity consumed, material produced, service provided, area of land affected.

3.2.2
primary data
quantified value of a process or an activity obtained from a direct measurement or a calculation based on direct measurements

Note 1 to entry: Primary data can include *GHG emission factors* (3.1.7) *or GHG removal factors* (3.1.8) and/or *GHG activity data* (3.2.1).

3.2.3
site-specific data
primary data (3.2.2) obtained within the *organizational boundary* (3.4.7)

Note 1 to entry: All site-specific data are primary data, but not all primary data are site-specific data.

3.2.4
secondary data
data obtained from sources other than *primary data* (3.2.2)

Note 1 to entry: Such sources can include databases and published literature validated by competent authorities.

3.2.5
greenhouse gas statement
GHG statement
DEPRECATED: GHG assertion
factual and objective declaration that provides the subject matter for the *verification* (3.4.9) or *validation* (3.4.10)

Note 1 to entry: The GHG statement could be presented at a point in time or could cover a period of time.

Note 2 to entry: The GHG statement provided by the *responsible party* (3.4.3) should be clearly identifiable, capable of consistent evaluation or measurement against suitable criteria by a *verifier* (3.4.11) or *validator* (3.4.12).

Note 3 to entry: The GHG statement could be provided in a *GHG report* (3.2.9) or *GHG project* (3.2.7) plan.

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3.2.6
greenhouse gas inventory
GHG inventory
list of *GHG sources* (3.1.2) and *GHG sinks* (3.1.3), and their quantified *GHG emissions* (3.1.5) and *GHG removals* (3.1.6)

3.2.7
greenhouse gas project
GHG project
activity or activities that alter the conditions of a GHG baseline and which cause *GHG emission* (3.1.5) reductions or *GHG removal* (3.1.6) enhancements

Note 1 to entry: ISO 14064-2 provides information on how to determine and use GHG baselines.

3.2.8
greenhouse gas programme
GHG programme
voluntary or mandatory international, national or subnational system or scheme that registers, accounts or manages *GHG emissions* (3.1.5), *GHG removals* (3.1.6), GHG emission reductions or GHG removal enhancements outside the *organization* (3.4.2) or *GHG project* (3.2.7)

3.2.9
greenhouse gas report
GHG report
standalone document intended to communicate an *organization's* (3.4.2) or *GHG project's* (3.2.7) GHG-related information to its *intended users* (3.4.4)

Note 1 to entry: A GHG report can include a *GHG statement* (3.2.5).

3.2.10
base year
specific, historical period identified for the purpose of comparing *GHG emissions* (3.1.5) or *GHG removals* (3.1.6) or other GHG-related information over time

3.2.11
greenhouse gas reduction initiative
GHG reduction initiative
specific activity or initiative, not organized as a *GHG project* (3.2.7), implemented by an *organization* (3.4.2) on a discrete or continuous basis, to reduce or prevent direct or indirect *GHG emissions* (3.1.5) or enhance direct or indirect *GHG removals* (3.1.6)

3.2.12
monitoring
continuous or periodic assessment of *GHG emissions* (3.1.5), *GHG removals* (3.1.6) or other GHG-related data

3.2.13
uncertainty
parameter associated with the result of quantification that characterizes the dispersion of the values that could be reasonably attributed to the quantified amount

Note 1 to entry: Uncertainty information typically specifies quantitative estimates of the likely dispersion of values and a qualitative description of the likely causes of the dispersion.

3.2.14
significant indirect greenhouse gas emission
significant indirect GHG emission
organization's (3.4.2) quantified and reported *GHG emissions* (3.1.5) complying with the significance criteria set by the organization

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3.3 Terms relating to biogenic material and land use

3.3.1
biomass
material of biological origin, excluding material embedded in geological formations and material transformed to fossilized material

Note 1 to entry: Biomass includes organic material (both living and dead), e.g. trees, crops, grasses, tree litter, algae, animals, manure and waste of biological origin.

3.3.2
biogenic carbon
carbon derived from *biomass* (3.3.1)

3.3.3
biogenic CO_2
CO_2 obtained by the oxidation of *biogenic carbon* (3.3.2)

3.3.4
anthropogenic biogenic GHG emission
GHG emission (3.1.5) from biogenic material as a result of human activities

3.3.5
direct land use change
dLUC
change in the human use of land within the relevant boundary

Note 1 to entry: Relevant boundary is the *reporting boundary* (3.4.8).

3.3.6
land use
human use or management of land within the relevant boundary

Note 1 to entry: Relevant boundary is the *reporting boundary* (3.4.8).

3.3.7
non-anthropogenic biogenic GHG emission
GHG emission (3.1.5) from biogenic material caused by natural disasters (e.g. wildfire or infestation by insects) or natural evolution (e.g. growth, decomposition)

3.4 Terms relating to organizations, interested parties and verification

3.4.1
facility
single installation, set of installations or production processes (stationary or mobile), which can be defined within a single geographical boundary, organizational unit or production process

3.4.2
organization
person or group of people that has its own functions with responsibilities, authorities and relationships to achieve its objectives

Note 1 to entry: The concept of organization includes, but is not limited to, sole-trader, company, corporation, firm, enterprise, authority, partnership, association, charity or institution, or part or combination thereof, whether incorporated or not, public or private.

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3.4.3
responsible party
person or persons responsible for the provision of the *GHG statement* (3.2.5) and the supporting *GHG* (3.1.1) ınformation

Note 1 to entry: The responsible party can be either individuals or representatives of an *organization* (3.4.2) or project, and can be the party who engages the *verifier* (3.4.11) or *validator* (3.4.12).

3.4.4
intended user
individual or *organization* (3.4.2) identified by those reporting GHG-related information as being the one who relies on that information to make decisions

Note 1 to entry: The intended user can be the *client* (3.4.5), the *responsible party* (3.4.3), the organization itself, *GHG programme* (3.2.8) administrators, regulators, the financial community or other affected interested parties, such as local communities, government departments, general public or non-governmental organizations.

3.4.5
client
organization (3.4.2) or person requesting *verification* (3.4.9) or *validation* (3.4.10)

3.4.6
intended use of the GHG inventory
main purpose set by the *organization* (3.4.2), or a programme, to quantify its *GHG emissions* (3.1.5) and *GHG removals* (3.1.6) consistent with the needs of the *intended user* (3.4.4)

3.4.7
organizational boundary
grouping of activities or facilities in which an *organization* (3.4.2) exercises operational or financial control or has an equity share

3.4.8
reporting boundary
grouping of *GHG emission* (3.1.5) or *GHG removals* (3.1.6) reported from within the *organizational boundary* (3.4.7), as well as those significant indirect emissions that are a consequence of the *organization's* (3.4.2) operations and activities

3.4.9
verification
process for evaluating a statement of historical data and information to determine if the statement is materially correct and conforms to criteria

3.4.10
validation
process for evaluating the reasonableness of the assumptions, limitations and methods that support a statement about the outcome of future activities

3.4.11
verifier
competent and impartial person with responsibility for performing and reporting on a *verification* (3.4.9)

3.4.12
validator
competent and impartial person with responsibility for performing and reporting on a *validation* (3.4.10)

3.4.13
level of assurance
degree of confidence in the *GHG statement* (3.2.5)

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4 Principles

4.1 General

The application of principles is fundamental to ensure that GHG-related information is a true and fair account. The principles are the basis for, and will guide the application of, the requirements in this document.

4.2 Relevance

Select the GHG sources, GHG sinks, GHG reservoirs, data and methodologies appropriate to the needs of the intended user.

4.3 Completeness

Include all relevant GHG emissions and removals.

4.4 Consistency

Enable meaningful comparisons in GHG-related information.

4.5 Accuracy

Reduce bias and uncertainties as far as is practical.

4.6 Transparency

Disclose sufficient and appropriate GHG-related information to allow intended users to make decisions with reasonable confidence.

5 GHG inventory boundaries

5.1 Organizational boundaries

The organization shall define its organizational boundaries.

The organization may comprise one or more facilities. Facility-level GHG emissions or removals may be produced from one or more GHG sources or sinks.

The organization shall consolidate its facility-level GHG emissions and removals by one of the following approaches:

a) control: the organization accounts for all GHG emissions and/or removals from facilities over which it has financial or operational control;

b) equity share: the organization accounts for its portion of GHG emissions and/or removals from respective facilities.

The consolidation approach shall be consistent with the intended use of the GHG inventory.

NOTE 1 Guidance on applying control and equity share approaches to consolidate facility-level GHG emissions and removals to the organization level is included in Annex A.

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The organization may use different consolidation approaches in the case of multiple reporting goals and requirements defined, for example, by the GHG programme, legal contract or different types of intended users.

NOTE 2 An organization's GHG emissions and removals are aggregated from facility-level quantification of GHG sources and sinks.

NOTE 3 A GHG sink in one period might become a GHG source in another period or vice versa.

When a facility is owned or controlled by several organizations, these organizations should adopt the same consolidation approach for that facility. The organization shall document and report which consolidation approach it applies.

5.2 Reporting boundaries

5.2.1 Establishing reporting boundaries

The organization shall establish and document its reporting boundaries, including the identification of direct and indirect GHG emissions and removals associated with the organization's operations.

5.2.2 Direct GHG emissions and removals

The organization shall quantify direct GHG emissions separately for CO_2, CH_4, N_2O, NF_3, SF_6 and other appropriate GHG groups (HFCs, PFCs, etc.) in tonnes of CO_2e.

The organization should quantify GHG removals.

5.2.3 Indirect GHG emissions

The organization shall apply and document a process to determine which indirect emissions to include in its GHG inventory.

As part of this process, the organization shall define and explain its own pre-determined criteria for significance of indirect emissions, considering the intended use of the GHG inventory.

Whatever the intended use is, criteria should not be used to exclude substantial quantities of indirect emissions or evade compliance obligations.

Using those criteria, the organization shall identify and evaluate its indirect GHG emissions, to select the significant ones.

The organization shall quantify and report these significant emissions. Exclusions of significant indirect emissions shall be justified.

The criteria to evaluate significance may include the magnitude/volume of the emissions, level of influence on sources/sinks, access to information and the level of accuracy of associated data (complexity of organization and monitoring). A risk assessment or other procedures (e.g. buyer requirements, regulatory requirements, concern of interested parties, scale of operation, etc.) may be used (see ISO 13065). More guidance is provided in Annex H.

The criteria for evaluating the significance may be periodically revised. The organization should retain documented information about the revisions.

5.2.4 GHG inventory categories

GHG emissions shall be aggregated into the following categories at the organizational level:

a) direct GHG emissions and removals;

b) indirect GHG emissions from imported energy;

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c) indirect GHG emissions from transportation;

d) indirect GHG emissions from products used by organization;

e) indirect GHG emissions associated with the use of products from the organization;

f) indirect GHG emissions from other sources.

In each category, non-biogenic emissions, biogenic anthropogenic emissions and, if quantified and reported, biogenic non-anthropogenic emissions shall be separated (see Annex D).

The organization should document the above categories separately at the facility level.

GHG emissions should be further subdivided into subcategories consistent with the above categories. An example of subcategories is provided in Annex B.

6 Quantification of GHG emissions and removals

6.1 Identification of GHG sources and sinks

The organization shall identify and document all relevant GHG sources and sinks included in its reporting boundaries. The organization shall include all relevant GHGs.

GHG sources and sinks shall be identified in accordance with the categories defined in 5.2.4.

If the organization quantifies GHG removals, the organization shall identify and document GHG sinks contributing to its GHG removals.

The detail with which sources and sinks are identified and categorized shall be consistent with the quantification approach used.

The organization may exclude GHG sources or sinks for which the contribution to GHG emissions or removals is not relevant. It shall identify and explain why the GHG sources or sinks are excluded in accordance with the categories and any categorical subdivisions included in the report (see 5.2.3).

6.2 Selection of quantification approach

6.2.1 General

The organization shall select and use quantification methodologies that minimize uncertainty and yield accurate, consistent and reproducible results.

The quantification approach should also consider technical feasibility and cost.

NOTE Quantification approach is the process of obtaining data and determining the emissions or removals from a source or sink. GHG emissions or removals can be obtained through measurement or modelling.

The organization shall explain and document its quantification approach and any changes in quantification approach.

6.2.2 Data selection and collection used for quantification

The organization shall identify and document its data for each source or sink classified as direct or indirect emissions and removals. It shall determine and document the characteristics for each relevant data used for quantification (see 5.2.3).

NOTE 1 Data used for quantification include primary data (including site specific) and secondary data.

EXAMPLE Data used for quantification may include the average of truck fuel consumption and its characteristics as the standard to determine fuel consumption.

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NOTE 2 In the case of GHG programmes, characteristics of data used for quantification are usually determined by the programme operator.

Annex C provides guidance on the selection and collection of data used for quantification.

6.2.3 Selection or development of GHG quantification model

Except in the case of measurement of emissions and removals, the organization shall select or develop models for the quantification approach.

A model is a representation of how the source or sink data used for quantification are converted into emissions or removals. A model is a simplification of physical processes that has assumptions and limitations.

The organization shall explain and document the justification for the selection or development of the model, considering the following model characteristics:

a) how the model accurately represents the emissions and removals;

b) its limits of application;

c) its uncertainty and rigour;

d) the reproducibility of results;

e) the acceptability of the model;

f) the origin and level of recognition of the model;

g) the consistency with the intended use.

NOTE Several types of models make use of activity data multiplied by emission factors.

6.3 Calculation of GHG emissions and removals

The organization shall calculate GHG emissions and removals in accordance with the quantification approach selected (see 6.2).

The period for which GHG emissions and removals have been calculated shall be reported.

The organization shall convert the quantity of each type of GHG to tonnes of CO_2e using appropriate GWPs.

The latest IPCC's GWP should be used. If not, justification shall be provided. The GWP time horizon shall be 100 years. Other GWP time horizons may be used, but reported separately.

NOTE GWP might be part of a model (including emissions factors).

The organization shall quantify biogenic emissions or removals in accordance with Annex D.

The organization shall quantify emissions or removals from imported electricity that is consumed by the organization, and of exported electricity generated by the organization, in accordance with Annex E.

Specific guidance concerning emissions or removals from agriculture is provided in Annex G.

6.4 Base-year GHG inventory

6.4.1 Selection and establishment of base year

The organization shall establish a historical base year for GHG emissions and removals for comparative purposes or to meet GHG programme requirements or other intended uses of the GHG inventory.

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Base-year emissions or removals may be quantified based on a specific period (e.g. a year or part of a year where seasonality is a feature of the organization's activity) or averaged from several periods (e.g. several years).

If sufficient information on historical GHG emissions or removals is not available, the organization may use its first GHG inventory period as the base year.

In establishing the base year, the organization:

a) shall quantify base-year GHG emissions and removals using data representative of the organization's current reporting boundary, typically single-year data, a consecutive multi-year average or a rolling average;

b) shall select a base year for which verifiable GHG emissions or removals data are available;

c) shall explain the selection of the base year;

d) shall develop a GHG inventory for the base year consistent with the provisions of this document.

The organization may change its base year, but shall justify any change to the base year.

6.4.2 Review of base-year GHG inventory

To ensure the representativeness of the base-year GHG inventory, the organization shall develop, document and apply a base-year review and recalculation procedure to account for substantial cumulative changes in base-year emissions resulting from:

a) a structural change in reporting or organizational boundaries (i.e. merger, acquisition or divestiture), or

b) a change in calculation methodologies or emission factors, or

c) the discovery of an error or a number of cumulative errors that are collectively substantial.

The organization shall not recalculate its base-year GHG inventory to account for changes in facility production levels, including the closing or opening of facilities.

The organization shall document base-year recalculations in subsequent GHG inventories.

7 Mitigation activities

7.1 GHG emission reduction and removal enhancement initiatives

The organization may plan and implement GHG reduction initiatives to reduce or prevent GHG emissions or enhance GHG removals.

If implemented, the organization should quantify GHG emission or removal differences attributable to the implementation of GHG reduction initiatives.

NOTE GHG emission or removal differences resulting from GHG reduction initiatives are usually reflected in the organization's GHG inventory, but can also result in GHG emission or removal differences outside GHG inventory boundaries.

If quantified and reported, the organization shall document GHG reduction initiatives and associated GHG emission or removal differences separately, and shall describe:

a) the GHG reduction initiatives;

b) the spatial and temporal boundaries of the GHG reduction initiatives;

c) the approach (appropriate indicators) used to quantify GHG emission or removal differences;

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d) the determination and classification of GHG emission or removal differences attributable to GHG reduction initiatives as direct or indirect GHG emissions or removals.

EXAMPLE GHG reduction initiatives might include the following:

— energy demand and use management;

— energy efficiency;

— technology or process improvements;

— GHG capture and storage in, typically, a GHG reservoir;

— management of transport and travel demands;

— fuel switching or substitution;

— afforestation;

— waste minimization;

— alternative fuels and raw materials (AFR) use to avoid landfilling or incinerating the wastes;

— refrigerant management.

7.2 GHG emission reduction or removal enhancement projects

If the organization reports offsets purchased or developed, the organization shall list such offsets separately from GHG reduction initiatives.

7.3 GHG emission reduction or removal enhancement targets

The organization may set targets to reduce GHG emissions.

If the organization reports a target, the following information shall be specified and reported:

— period covered by the target, including the target reference year and the target completion year;

— type of target (intensity or absolute);

— category of emissions included in the target;

— the amount of reduction and its unit expressed in accordance with the type of target.

For setting the target, the following criteria should be considered:

— climate science;

— reduction potential;

— international, national context;

— sectorial context (e.g. voluntary sectorial commitment, cross-sectorial effect).

8 GHG inventory quality management

8.1 GHG information management

8.1.1 The organization shall establish and maintain GHG information management procedures that:

a) ensure conformity with the principles of this document;

b) ensure consistency with the intended use of the GHG inventory;

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c) provide routine and consistent checks to ensure accuracy and completeness of the GHG inventory;

d) identify and address errors and omissions;

e) document and archive relevant GHG inventory records, including information management activities and GWPs.

8.1.2 The organization's GHG information management procedures shall document their consideration of the following:

a) identification and review of the responsibility and authority of those responsible for GHG inventory development;

b) identification, implementation and review of appropriate training for members of the inventory development team;

c) identification and review of organizational boundaries;

d) identification and review of GHG sources and sinks;

e) selection and review of quantification approaches, including data used for quantification and GHG quantification models that are consistent with the intended use of the GHG inventory;

f) review of the application of quantification approaches to ensure consistency across multiple facilities;

g) use, maintenance and calibration of measurement equipment (if applicable);

h) development and maintenance of a robust data-collection system;

i) regular accuracy checks;

j) periodic internal audits and technical reviews;

k) periodic review of opportunities to improve information management processes.

8.2 Document retention and record keeping

The organization shall establish and maintain procedures for document retention and record keeping.

The organization shall retain and maintain documentation supporting the design, development and maintenance of the GHG inventory to enable verification. The documentation, whether in paper, electronic or other format, shall be handled in accordance with the organization's GHG information management procedures for document retention and record keeping.

8.3 Assessing uncertainty

The organization shall assess the uncertainty associated with the quantification approaches (e.g. data used for quantification and models) and conduct an assessment that determines the uncertainty at the GHG inventory category level.

Where quantitative estimation of uncertainty is not possible or cost effective, it shall be justified and a qualitative assessment shall be conducted.

The organization may apply the principles and methodologies of ISO/IEC Guide 98-3 in completing the uncertainty assessment.

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9 GHG reporting

9.1 General

The organization should prepare a GHG report, consistent with the intended uses of the GHG inventory, to facilitate GHG inventory verification. For example, a GHG report may be necessary for participation in a GHG programme or to inform external or internal users.

A GHG report shall be prepared if the organization chooses to have its GHG inventory verified or makes a public GHG statement claiming conformity with this document.

GHG reports shall be complete, consistent, accurate, relevant, transparent and planned in accordance with 9.2.

If the organization's GHG statement has been independently (third-party) verified, the verification statement shall be made available to intended users.

If confidential data are withheld from inclusion in a GHG report, this shall be justified.

If the organization decides to prepare a GHG report, 9.2 and 9.3 apply.

9.2 Planning the GHG report

The organization shall explain and document the following in planning its GHG report:

a) purpose and objectives of the report in the context of the organization's GHG policies, strategies or programmes, and applicable GHG programmes;

b) intended use and intended users of the GHG inventory;

c) overall and specific responsibilities for preparing and producing the report;

d) frequency of the report;

e) report structure and format;

f) data and information to be included in the report;

g) policy on availability and methods of dissemination of the report.

9.3 GHG report content

9.3.1 Required information

The organization's GHG report shall describe the organization's GHG inventory. Its content may be structured as recommended in Annex F.

GHG report content shall include the following:

a) description of the reporting organization;

b) person or entity responsible for the report;

c) reporting period covered;

d) documentation of organizational boundaries (5.1);

e) documentation of reporting boundaries, including criteria determined by the organization to define significant emissions;

f) direct GHG emissions, quantified separately for CO_2, CH_4, N_2O, NF_3, SF_6 and other appropriate GHG groups (HFCs, PFCs, etc.) in tonnes of CO_2e (5.2.2);

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g) a description of how biogenic CO_2 emissions and removals are treated in the GHG inventory and the relevant biogenic CO_2 emissions and removals quantified separately in tonnes of CO_2e (see Annex D);

h) if quantified, direct GHG removals, in tonnes of CO_2e (5.2.2);

i) explanation of the exclusion of any significant GHG sources or sinks from the quantification (5.2.3);

j) quantified indirect GHG emissions separated by category in tonnes of CO_2e (5.2.4);

k) the historical base year selected and the base-year GHG inventory (6.4.1);

l) explanation of any change to the base year or other historical GHG data or categorization and any recalculation of the base year or other historical GHG inventory (6.4.1), and documentation of any limitations to comparability resulting from such recalculation;

m) reference to, or description of, quantification approaches, including reasons for their selection (6.2);

n) explanation of any change to quantification approaches previously used (6.2);

o) reference to, or documentation of, GHG emission or removal factors used (6.2);

p) description of the impact of uncertainties on the accuracy of the GHG emissions and removals data per category (8.3);

q) uncertainty assessment description and results (8.3);

r) a statement that the GHG report has been prepared in accordance with this document;

s) a disclosure describing whether the GHG inventory, report or statement has been verified, including the type of verification and level of assurance achieved;

t) the GWP values used in the calculation, as well as their source. If the GWP values are not taken from the latest IPCC report, include the emissions factors or the database reference used in the calculation, as well as their source.

9.3.2 Recommended information

The organization should consider including in the GHG report:

a) description of the organization's GHG policies, strategies or programmes;

b) if appropriate, description of GHG reduction initiatives and how they contribute to GHG emission or removal differences, including those occurring outside organizational boundaries, quantified in tonnes of CO_2e (7.1);

c) if appropriate, purchased or developed GHG emission reductions and removal enhancements from GHG emission reduction and removal enhancement projects, quantified in tonnes of CO_2e (7.2);

d) as appropriate, description of applicable GHG programme requirements;

e) GHG emissions or removals disaggregated by the facility;

f) total quantified indirect GHG emissions;

g) description and presentation of additional indicators, such as efficiency or GHG emission intensity (emissions per unit of production) ratios;

h) assessment of performance against appropriate internal and/or external benchmarks;

i) description of GHG information management and monitoring procedures (8.1);

j) GHG emissions and removals from the previous reporting period;

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k) if appropriate, explanation of GHG emissions differences between the present inventory and the previous one.

The organization may aggregate direct emissions and direct removals.

9.3.3 Optional information and associated requirements

The organization may report optional information separately from the required information and the recommended information. Each type of optional information described below should be reported separately from the others.

The organization may report the results of contractual instruments for GHG attributes (market based approach), expressed in GHG emissions (tCO_2e) as well as in the unit of transfer (e.g. kWh). The organization may report the amount purchased compared to the amount consumed.

The organization may report offsets or other types of carbon credits. If so, the organization:

shall disclose the GHG scheme under which they were generated;

may add offsets or other types of carbon credits together if they originate from the same GHG scheme and are of appropriate vintage;

shall not add or subtract offsets or other types of carbon credits from the organization's inventory of its direct or indirect emissions.

The organization may report GHGs stored in GHG reservoirs.

10 Organization's role in verification activities

The organization may decide to conduct a verification.

To review GHG emissions and removals information, impartially and objectively, the organization shall conduct a verification consistent with the needs of the intended user. Principles and requirements are described in ISO 14064-3.

Requirements for verification bodies are described in ISO 14065.

Requirements for the competence of validation teams and verification teams are described in ISO 14066.

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Annex A
(informative)

Process to consolidate data

A.1 General

Organizations are encouraged to refer to ISO/TR 14069 for additional guidance in applying consolidation approaches and dealing with double counting.

An organization engaged in setting its organizational boundaries should first define the intended uses of its GHG inventory by considering its GHG policies, strategies or programmes, operations and facilities to determine the GHG sources which it can control and those which it may influence.

The intended use of the inventory can help to determine the organizational boundaries (see H.1). In developing its GHG quantification and reporting system, an organization should ensure that the data system can meet a range of reporting requirements. GHG data should be recorded and quantified by source, sink and type, at least to the facility level. Such data should be retained in its disaggregated form to provide maximum flexibility in meeting a range of reporting requirements. Consolidation of the information can then be carried out as required.

If the GHG emissions and removals are quantified at the facility level, and according to the intended use of the GHG inventory, one of the two approaches outlined in A.2 and A.3 should be selected to guide and assist in the consolidation of facility data to the organization level.

Where possible, organizations should follow the organizational boundaries already in place for their financial accounting, provided these are explicitly explained and followed consistently. When applying these concepts, the underlying assumption of "substance over form" should be followed. That is, GHG emissions and removals should be quantified and reported in accordance with the organization's substance and economic reality and not merely its legal form.

A.2 Consolidation based on control

Under the control approach, an organization accounts for 100 % of the GHG emissions or removals from operations over which it has control. It does not account for GHG emissions or removals from operations in which it owns an interest but has no control. Control may be defined in either financial or operational terms. When using the control approach to consolidate GHG emissions or removals, organizations may choose between either the operational control or financial control criteria.

An organization has financial control over the operation if it has the ability to direct the financial and operating policies of the operation with a view to gaining economic benefits from its activities. An organization has operational control over an operation if it, or one of its subsidiaries, has the full authority to introduce and implement its operating policies at the operational level.

A.3 Consolidation based on equity share

Equity share is the percentage of economic interest in, or benefit derived from, a facility. This consolidation approach increases the usability of GHG information for different users, and aims to mirror as far as possible the approach adopted by financial accounting and reporting standards. The equity share approach can be particularly useful for multinational companies with operations in a number of different jurisdictions aiming to determine their GHG inventory.

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Consolidating to the organizational level based on equity share requires establishing the ownership percentage of each facility, and accounting for that percentage of GHG emissions or removals from respective facilities, including using production share agreements.

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